RESEARCH PROJECT PORTFOLIO

AMR

Adaptive Multi-Expert Reasoning

via Difficulty-Aware Routing and Uncertainty-Guided Aggregation

Mohamed Ehab Mohamed

MSA University • Machine Learning Researcher • LLM & Reasoning Specialist

Under Review • JCSSE 2026 | **75.28% on GSM8K — No Synthetic Data** | **Outperforms MetaMath, WizardMath & ToRA**

01 | PROJECT OVERVIEW

AMR (Adaptive Multi-Expert Reasoning) is a novel LLM inference framework designed to solve a fundamental limitation of large language models: inconsistent reasoning performance across problems of varying difficulty. Rather than applying the same strategy to every problem, AMR dynamically adapts its reasoning approach based on predicted difficulty and uncertainty — selecting the right experts, generating the right number of candidates, and aggregating answers intelligently.

Evaluated on the GSM8K mathematical reasoning benchmark (1,319 test examples), AMR achieved 75.28% accuracy using only the original training data — with no synthetic data augmentation. This result outperforms the majority of 7B-scale models that relied on hundreds of thousands of synthetic examples, demonstrating that smart inference-time architecture can rival expensive data scaling strategies.

02 | PROJECT DETAILS

Project Type	Research Paper — Novel LLM Inference Framework for Mathematical Reasoning
Domain	Large Language Models • Mathematical Reasoning • Mixture of Experts • Uncertainty Estimation
Publication Status	Under Review — JCSSE 2025 (International Joint Conference on Computer Science and Software Engineering)
Benchmark	GSM8K Test Split — 1,319 Mathematical Word Problems
Base Model	Qwen2.5-Math 7B with LoRA-adapted Expert Modules
Core Technologies	Python • PyTorch • HuggingFace Transformers • LoRA • DeBERTa-v3 • Clustering Algorithms
Affiliation	MSA University — Faculty of Computer Science, Giza, Egypt

03 | KEY RESULTS & METRICS

75.28%	82.6%	64.1%	1,319
GSM8K Accuracy	Easy Problems	Hard Problems	Test Examples
No synthetic data used	Predicted easy split	Predicted hard split	GSM8K test split

AMR Accuracy by Difficulty Split

Difficulty Split	Correct	Total	Accuracy
Easy (Predicted)	**657**	**795**	**82.6%**
Hard (Predicted)	336	524	64.1%
Easy (Gold Standard)	**636**	**742**	**85.7%**
Hard (Gold Standard)	357	577	61.9%
Overall	**993**	**1,319**	**75.28%**

Comparison with State-of-the-Art 7B Models on GSM8K

Model	Size	Synthetic Data	GSM8K Accuracy
AMR (Proposed)	**7B**	**None ✓**	**75.28%**
MetaMath	7B	MetaMathQA (395K)	66.7%
MetaMath	13B	MetaMathQA (395K)	70.8%
MetaMath	70B	MetaMathQA (395K)	82.1%
MetaMath-Mistral	7B	MetaMathQA (395K)	77.8%
WizardMath	7B	Evol-Instruct	54.9%
WizardMath	13B	Evol-Instruct	63.9%
ToRA-Code	7B	TORA-CORPUS (16K)	72.6%
ToRA-Code	13B	TORA-CORPUS (16K)	75.8%
ToRA-Code	34B	TORA-CORPUS (16K)	80.7%
ToRA-Code	70B	TORA-CORPUS (16K)	84.3%
OVM(LLaMA-2)	7B+7B	Additional	73.7%
OVM(Mistral)	7B+7B	Additional	84.7%
MAMmoTH	7B	MathInstruct (260K)	52.8%
MAMmoTH	13B	MathInstruct (260K)	62.4%
MAMmoTH	70B	MathInstruct (260K)	75.9%
MAMmoTH-Coder	7B	MathInstruct (260K)	59.9%
MAMmoTH-Coder	13B	MathInstruct (260K)	64.9%

SEGO	7B	GSM8K+MATH+AQuA	68.7%
SEGO	13B	GSM8K+MATH+AQuA	72.5%
Abel	7B	Unreleased	59.5%
Abel	13B	Unreleased	66.7%
Abel	70B	Unreleased	83.9%
Phi-GSM - Phi-1.5	1.3B	TinyGSM (1.3M)	68.2%
Phi-GSM+V - Phi-1.5	1.3B+1.3B	TinyGSM (1.3M)	81.5%
Phi-GSM (125M) - Phi-1.5-tiny	125M	TinyGSM (1.3M)	63.1%
Phi-GSM+V (125M) - Phi-1.5-tiny	125M+125M	TinyGSM (1.3M)	68.9%
Phi-GSM (350M) - Phi-1.5-small	350M	TinyGSM (1.3M)	65.9%
Phi-GSM+V (350M) - Phi-1.5-small	350M+350M	TinyGSM (1.3M)	71.3%
Phi-GSM - Phi-2	2.7B	TinyGSM (1.3M)	74.3%

04 | TECHNICAL APPROACH

Framework Overview — 4 Core Components

① Difficulty-Aware Router

Predicts problem difficulty (easy/hard) and computes a hybrid uncertainty score using Shannon entropy and margin formulation. Controls generation diversity across 3 regimes: Low ($U < 0.35$) deterministic, Medium (0.35–0.55) single candidate, High (≥ 0.55) two diverse candidates per expert.

② Multi-Expert Reasoning System

Three LoRA-adapted specialist experts each trained with different reasoning styles: Algebraic (equation-based), Intuitive (mental math/natural language), and Step-by-Step (structured line-by-line derivations). Includes correction and finalization passes to improve answer quality.

③ Neural Verifier

A DeBERTa-v3 binary classifier trained on problem-solution pairs. Assigns a correctness confidence score between 0 and 1 for each candidate answer. Provides reliable quality estimates that feed into the final aggregation scoring formula.

④ Clustering-Based Aggregation

Groups candidates by numerical answer and scores each cluster using a weighted formula combining verifier confidence (50%), completion quality (18%), answer coherence (16%), and source bonus (16%). Selects the best candidate from the highest-scoring cluster.

Key Innovation — What Makes AMR Different

- Data efficiency: Achieves 75.28% on GSM8K using only original training data — no synthetic augmentation
- Adaptive inference: Dynamically adjusts reasoning strategy per problem rather than applying uniform prompting
- Multi-style diversity: Three stylistically distinct LoRA experts generate complementary solution candidates
- Uncertainty modeling: Hybrid entropy-margin formulation controls generation diversity adaptively
- Verifier-guided selection: Neural correctness scoring combined with clustering ensures reliable final answers

05 | KEY CONTRIBUTIONS & IMPACT

New Inference-Time Dimension

Prior work improved LLM reasoning through robustness benchmarking, data scaling, or prompt engineering. AMR introduces a fourth dimension: inference-time architecture — demonstrating that routing and aggregation alone can achieve competitive accuracy.

Data Efficiency Benchmark

AMR is the first framework to achieve 75.28% on GSM8K using only the original training set, surpassing MetaMath-7B (66.7%), WizardMath-7B (54.9%), and ToRA-7B (72.6%) — all of which relied on synthetic datasets of 16K–395K examples.

Difficulty-Sensitive Routing

The router achieves 73.4% alignment with gold difficulty labels at test time — using only the problem text as input. This enables meaningful differentiation between easy and hard problems, directly improving robustness on the hard subset (64.1% vs single-run baselines).

Orthogonal to Existing Methods

AMR's inference-time architecture is fully complementary to prompt engineering methods like DUP and data scaling methods like TinyGSM. When combined, these approaches have strong potential to push performance significantly beyond current state-of-the-art.

06 | ABOUT THE RESEARCHER

Mohamed Ehab Mohamed

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Education: Dual B.Sc. Computer Science — MSA University (Egypt) & University of Greenwich (UK), Expected 2026

Specialization: NLP, LLMs, Mathematical Reasoning, Mixture of Experts, Deep Learning, End-to-End ML Pipelines

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